

Hospital Management System

1. Find all patients whose treatment cost is greater than 20,000 and display only their name and department

```
> db.patients.aggregate([
  {$match: { treatmentCost: { $gt: 20000 } }},
  {$project: {_id: 0,patientName: 1,department: 1}}
])
< {
  patientName: 'Meena',
  department: 'Neurology'
}
{
  patientName: 'Kiran',
  department: 'Neurology'
}
{
  patientName: 'Vijay',
  department: 'Cardiology'
}
{
  patientName: 'Priya',
```

2. List all patients sorted by treatment cost in descending order.

```
> db.patients.aggregate([
  {$sort:{treatmentCost:-1}}
])
< {
  _id: ObjectId('6a1336eb94785527f5c440fd'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
{
  _id: ObjectId('6a1336eb94785527f5c44103'),
  patientName: 'Priya',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 48,
```

3. Display the first 5 patient records.

```
> db.patients.aggregate([{$limit:5}])
< {
  _id: ObjectId('6a1336eb94785527f5c440fc'),
  patientName: 'Arun',
  doctor: 'Dr. Ravi',
  department: 'Cardiology',
  age: 45,
  treatmentCost: 20000,
  daysAdmitted: 5
}
{
  _id: ObjectId('6a1336eb94785527f5c440fd'),
  patientName: 'Meena',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 50,
  treatmentCost: 30000,
  daysAdmitted: 7
}
```

4. Skip first 3 patients and display the remaining records.

```
    daysAdmitted: 6
  }
db.patients.aggregate([{$skip:3}])
{
  _id: ObjectId('6a1336eb94785527f5c440ff'),
  patientName: 'Divya',
  doctor: 'Dr. Mehta',
  department: 'Orthopedics',
  age: 35,
  treatmentCost: 10000,
  daysAdmitted: 2
}
{
  _id: ObjectId('6a1336eb94785527f5c44100'),
  patientName: 'Kiran',
  doctor: 'Dr. Kumar',
  department: 'Neurology',
  age: 40,
```

5. Find patients from the Cardiology department and display only their names

```
> db.patients.aggregate([
  {$match: { department:'Cardiology' }},
  {$project: {_id: 0,patientName: 1,}}
])
< {
  patientName: 'Arun'
}
{
  patientName: 'Raj'
}
{
  patientName: 'Vijay'
}
{
  patientName: 'Riya'
}
```

6. Find total treatment cost per department.

```
> db.patients.aggregate([
  {$group: {_id: "$department",totalCost: { $sum: "$treatmentCost" }}}
])
< {
  _id: 'Orthopedics',
  totalCost: 30000
}
{
  _id: 'Cardiology',
  totalCost: 75000
}
{
  _id: 'Neurology',
  totalCost: 82000
}
```

7. Find average treatment cost per doctor and display results in ascending order.

```
> db.patients.aggregate([
  {$group: {_id: "$doctor", avgCost: { $avg: "$treatmentCost" }}}},
  {$sort : {avgCost:1}}
])
< {
  _id: 'Dr. Mehta',
  avgCost: 10000
}
{
  _id: 'Dr. Ravi',
  avgCost: 18750
}
{
  _id: 'Dr. Kumar',
  avgCost: 27333.333333333332
}
```

8. Find departments where total treatment cost is greater than 50,000.

```
> db.patients.aggregate([
  {$group: {_id: "$department", totalCost: { $sum: "$treatmentCost" }}}},
  {$match:{totalCost:{$gt:50000}}}
])
< {
  _id: 'Cardiology',
  totalCost: 75000
}
{
  _id: 'Neurology',
  totalCost: 82000
}
```

9. Find number of patients in each department and sort the result in descending order

```

    }
  > db.patients.aggregate([
    {$group: {_id: "$department",patientCount: { $sum: 1 }}},
    {$sort: { patientCount: -1 }}
  ])
< {
  _id: 'Cardiology',
  patientCount: 4
}
{
  _id: 'Neurology',
  patientCount: 3
}
{
  _id: 'Orthopedics',
  patientCount: 3
}
}

```

10. Find top 3 doctors with highest total treatment cost.

```

> db.patients.aggregate([
  {$group: {_id: "$doctor",totalCost: { $sum: "$treatmentCost" }}},
  {$sort : {totalCost:-1}},
  {$limit:3}
])
< {
  _id: 'Dr. Kumar',
  totalCost: 82000
}
{
  _id: 'Dr. Ravi',
  totalCost: 75000
}
{
  _id: 'Dr. Mehta',
  totalCost: 30000
}
}

```

11. Find patients whose age is greater than 40, group them by department, and count number of patients.

```
> db.patients.aggregate([
  {$match:{age:{$gt:40}}},
  {$group: {_id:"$department",patientCount:{$sum:1}}}
])
< {
  _id: 'Orthopedics',
  patientCount: 1
}
{
  _id: 'Cardiology',
  patientCount: 3
}
{
  _id: 'Neurology',
  patientCount: 2
}
```


12. Find average treatment cost per department, display only department and average cost, and sort descending.

```
db.patients.aggregate([
  {$group: {_id: "$department", averageCost: {$avg: "$treatmentCost"}}},
  {$project: {_id: 0, department: "$_id", averageCost: 1}},
  {$sort: {averageCost: -1}}
])
[
  {
    averageCost: 27333.333333333332,
    department: 'Neurology'
  }
  {
    averageCost: 18750,
    department: 'Cardiology'
  }
  {
    averageCost: 10000,
    department: 'Orthopedics'
  }
]
```

13. Find departments having more than 2 patients admitted.

```
> db.patients.aggregate([
  {$group: {_id: "$department", patientcount: {$sum: 1}}},
  {$match: {patientcount: {$gt: 2}}}
])
< {
  _id: 'Neurology',
  patientcount: 3
}
{
  _id: 'Cardiology',
  patientcount: 4
}
{
  _id: 'Orthopedics',
  patientcount: 3
}
```

14. Display top 3 departments based on highest total treatment cost after skipping the first one.

```
> db.patients.aggregate([
  {$group: {_id: "$department", totalcost: {$sum: '$treatmentCost'}}},
  {$sort: {department: -1}},
  {$skip: 1},
  {$limit: 3}
])
< {
  _id: 'Cardiology',
  totalcost: 75000
}
{
  _id: 'Neurology',
  totalcost: 82000
}
```

15. Find doctors treating patients with cost greater than 10,000, calculate total cost, and show only top 2 doctors.

```
> db.patients.aggregate([
  {$match:{treatmentCost:{$gt:10000}}},
  {$group:{_id:"$doctor",totalcost:{$sum:'$treatmentCost'}}},
  {$sort:{totalcost:-1}},
  {$limit:2}
])
< {
  _id: 'Dr. Kumar',
  totalcost: 82000
}
{
  _id: 'Dr. Ravi',
  totalcost: 75000
}
```